

# PQC H-365

## High-Performance Quantum-Resistant Hybrid VPN & Multi-Message Secure Channel

NIST FIPS 203 (ML-KEM-768) + NIST FIPS 204 (ML-DSA-65) with Asynchronous Parallel Handshake Acceleration and 64-bit Monotonic Replay-Protected Multi-Message Transport.

### TEAM: PQC H-365

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<https://github.com/chipashacosmas/PQC-H365>

Engineered in C with Open Quantum Safe (liboqs)

# 1. The Looming Quantum Threat to VPNs & Key Exchange

## The Quantum Break

- Shor's Algorithm on Cryptanalytically Relevant Quantum Computers (CRQCs) solves discrete logs and factoring in polynomial time.
- RSA-2048, Diffie-Hellman, and ECDH (P-256 / X25519) will be fully compromised.
- Current enterprise VPN tunnels and TLS links will provide zero forward secrecy.

## Harvest Now, Decrypt Later

- Hostile nation-states and adversaries are recording encrypted network communications today.
- Encrypted intellectual property, defense cables, and financial data are retained in bulk.
- Once quantum hardware arrives, stored archives will be retroactively decrypted.
- Migration must happen now — before quantum computers mature.

## The Engineering Bottleneck

- Post-quantum algorithms produce kilobyte-scale keys and signatures (1KB to 3KB vs 32B).
- Serial hybrid handshakes compound algorithmic execution, causing 50+ ms stalls.
- Packets exceed standard network MTU (1500B), causing fragmentation and packet loss.
- Production VPNs need a line-rate, low-latency PQC architecture.

## 2. PQC H-365 High-Performance Hybrid Architecture

### Dual-Security Hybrid Key Exchange

- Combines classical X25519 (ECDH) with NIST FIPS 203 (ML-KEM-768).
- Shared secrets are fused via RFC 5869 HKDF-SHA256 into a 32-byte session key.
- Dual-Security: Adversaries must break BOTH elliptic curves and lattice mathematics simultaneously.

### Mutual Post-Quantum Auth (mPQC)

- Implements NIST FIPS 204 (ML-DSA-65) post-quantum digital signatures.
- Server and client sign the entire ephemeral handshake transcript.
- Eliminates active quantum Man-in-the-Middle (MitM) and quantum downgrade attacks.

### Parallel Handshake Scheduling

- POSIX multithreaded asynchronous key exchange pipeline in native C.
- Computes classical X25519 and ML-KEM-768 concurrently.
- Reduces handshake latency by 81.5% on client and 79.1% on server (down to ~11ms).

### Replay-Protected Secure Channel

- Persistent multi-message AES-256-GCM authenticated transport.
- 64-bit monotonic sequence numbers bound directly to AES-GCM nonces.
- Hardware wire-speed encryption with authenticated clean session teardown.

### 3. Standards Compliance & Cryptographic Primitives

| Cryptographic Layer     | Algorithm / Standard          | Public Key Size | Ciphertext / Signature | Role in Protocol                         |
|-------------------------|-------------------------------|-----------------|------------------------|------------------------------------------|
| Post-Quantum KEM        | NIST FIPS 203 (ML-KEM-768)    | 1,184 Bytes     | 1,088 Bytes            | Quantum-safe key encapsulation           |
| Post-Quantum Signature  | NIST FIPS 204 (ML-DSA-65)     | 1,952 Bytes     | 3,309 Bytes            | Mutual transcript authentication (mPQC)  |
| Classical Ephemeral DH  | X25519 (RFC 7748)             | 32 Bytes        | 32 Bytes               | Classical backward compatibility & speed |
| Key Derivation Function | HKDF-SHA256 (RFC 5869)        | N/A             | 32 Bytes               | Hybrid session key fusion                |
| Authenticated Transport | AES-256-GCM (NIST SP 800-38D) | N/A             | 16 Byte Tag            | Replay-protected multi-message tunnel    |

CNSA 2.0 & FIPS Compliance Mandate

Aligns strictly with NSA Commercial National Security Algorithm Suite 2.0 (CNSA 2.0) deployment timeline and NIST standardized post-quantum specifications using Open Quantum Safe (liboqs 0.10.0) in high-efficiency C.

## 4. Core Innovation: Parallel Handshake Scheduling

### CLIENT HANDSHAKE LATENCY

**81.5% Reduction**

**55.56 ms → 10.27 ms**

Concurrent X25519 key generation and ML-KEM-768 encapsulation executed via POSIX worker threads, bounding cryptographic execution time to  $\max(T_{\text{classical}}, T_{\text{pqc}})$ .

### SERVER HANDSHAKE LATENCY

**79.1% Reduction**

**56.31 ms → 11.79 ms**

Server decapsulates ML-KEM-768 while simultaneously computing X25519 shared secret, eliminating serial algorithmic round-trip stalls.



### Production Engineering Impact

- **Sequential Overhead Eliminated:** Traditional serial hybrid handshakes exceed 110ms total round-trip time, making high-concurrency enterprise gateways unresponsive.
- **Line-Rate Tunnel Establishment:** PQC H-365 achieves a ~11ms full hybrid handshake, matching classical VPN response times with complete quantum safety.
- **Hardware Scalability:** Low CPU latency enables commodity cloud instances to sustain thousands of authenticated handshakes per second.

## 5. System Design: Adaptive Framing & Anti-Replay Defense

### Adaptive Framing Engine

- **Zero-Copy Length-Prefixed Framing:** 4-byte network byte order prefix encodes payload length explicitly.
- **Dynamic Token Accommodation:** Handles 1,184B KEM public keys and 3,309B DSA signatures seamlessly.
- **Strict Boundary Enforcement:** Maximum frame cap at 4,096 bytes; oversized frames (>4096B) rejected immediately.
- **Memory Safety:** Malformed packets rejected before reaching cryptographic routines, eliminating buffer-overflow and DoS vectors.

### 64-bit Monotonic Replay Protection

- **Monotonic Counter Nonce:** 64-bit sequence counter prepended to every frame and embedded into GCM nonce.
- **Wire Format:** [8B Counter | 16B GCM Tag | Ciphertext]
- **AAD Binding:** Authenticated Additional Data binds string "PQC\_H365\_CHANNEL\_V1".
- **Replay Rejection:** Server maintains last seen counter; strictly drops any packet where `counter ≤ last_seen`.
- **Clean Teardown:** Authenticated `QUIT` frame initiates mutual socket closure.

# 6. Empirical Benchmark Results (20-Run Distribution)

| Handshake Mode              | Client Avg (ms) | Server Avg (ms) | Total Latency (ms) | Quantitative Impact                   |
|-----------------------------|-----------------|-----------------|--------------------|---------------------------------------|
| Classical X25519 Baseline   | 1.558 ms        | 0.547 ms        | 2.105 ms           | Baseline (Insecure vs Quantum)        |
| ML-KEM-768 Standalone       | 8.432 ms        | 7.225 ms        | 15.657 ms          | PQC Only (No Classical Fallback)      |
| Sequential Hybrid           | 55.559 ms       | 56.314 ms       | 111.873 ms         | Serial Overhead (Slow)                |
| Parallel Hybrid (PQC H-365) | 10.268 ms       | 11.789 ms       | 22.057 ms          | 81.5% Client / 79.1% Server Reduction |



## Rigorous Empirical Methodology

- Measured across 20 automated iterations over Linux network sockets using high-precision POSIX `clock_gettime()` counters.
- Raw reproducibility datasets committed and published on GitHub: [benchmarks/handshake\\_results.csv](#) and [benchmarks/handshake\\_summary.md](#).

## 7. Working Prototype & Live Verification Proof

### ✓ Automated Test Suite (100% Pass)

- **Algorithm Verification:** `check_liboqs` validates ML-KEM-768 & ML-DSA-65 runtime.
- **Known Answer Tests (KAT):** `test_ml_kem`, `test_ml_dsa`, `test_x25519`, and `test_hybrid_kdf` all PASS.
- **Adaptive Buffer Test:** Frame transfers for 1184B, 3309B, 4096B pass; 4097B rejected as expected.
- **Regression Runner:** `./scripts/run_all_tests.sh` executes complete suite with zero errors.

### 📄 Live Channel Execution Evidence

- **Handshake:** Server ML-DSA-65 signature verified: `[CLIENT] server sig: PASS`.
- **Key Agreement:** Both derive identical session key SHA256 prefix: `59b7b6397a0ea3bd`.
- **Payload 1 (ctr=1):** Encrypted with AES-GCM; server echoes `ECHO: hello PQC SECURE CHANNEL`.
- **Payload 2 (ctr=2):** Monotonic counter increment verified; message decrypted and echoed.
- **Teardown (ctr=3):** Clean shutdown triggered via `QUIT`; session closed gracefully.

## 8. Industry Applications & National Security Impact



### Defense & Intelligence

- Protects critical military communications against foreign state-sponsored HNDL intercept programs.
- Dual-Security guarantee eliminates single-algorithm failure risk.
- Pre-empts CNSA 2.0 government compliance deadlines.



### Banking & Finance

- Secures SWIFT inter-bank financial transactions and long-lived mortgage contracts.
- Eliminates latency penalties on high-frequency trades (~11ms handshake).
- Strict anti-replay protection prevents fraudulent packet duplication.



### Enterprise Cloud VPN

- Drop-in quantum-safe upgrade for enterprise WireGuard and OpenVPN gateways.
- Memory-safe C implementation with zero-copy buffer architecture.
- Low CPU overhead allows thousands of active client tunnels per node.

## 9. Roadmap: Round 2 Sprint & Grand Finale

### Round 2 Enhancement Sprint (Oct 10 – 29)

- **Linux TUN/TAP Driver (Step 29):** Binding `tun0` virtual interface to route real OS-level IPv4/IPv6 packet flows.
- **eBPF / XDP In-Kernel Acceleration:** Implementing bare-metal packet filtering and line-rate decryption hooks.
- **Multi-Path Interface Bonding:** AEAD frame multiplexing across dual network physical interfaces for link failover.
- **Adversarial Fuzzing:** Scapy/Kali fuzzing suite validating resilience against malformed quantum frames.

### Grand Finale (Bengaluru: Oct 30 – 31)

- **Live Hardware Demonstration:** Operating PQC H-365 tunnel on physical machines with Wireshark packet analysis.
- **Live Quantum MitM Attack Simulation:** Demonstrating active quantum stripping rejection and signature failure.
- **Telemetry Control Plane:** Demonstrating live latency charts and active client tunnel telemetry.
- **Open Source Release:** Packaged deployment binaries for Ubuntu / Debian and Docker containers.

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